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LABORATORY REPORT

OPERATING SYSTEMS LABORATORY 5

The File System with Your Own Hands

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ABSTRACT

This laboratory report investigates the Linux Virtual Filesystem (VFS) architecture through direct experimentation on an Ubuntu 24.04 LTS host running Linux kernel 6.6.87.2 on x86_64. All tasks were performed under an unprivileged user (`ubuntu`, UID 1000) to ensure Discretionary Access Control (DAC) evaluations were genuinely tested. The study covers namespace organization, inode metadata separation from directory entries, hard link versus symbolic link mechanics, physical disk allocation of sparse files, directory traversal versus enumeration permissions, setuid privilege elevation, and the pseudo-filesystem interface under `/proc`. In addition, all four optional stretch exercises are examined: binary magic number identification, setuid binary discovery, active file descriptor unlinking, and loopback ext4 filesystem creation with inode quota verification.

Keywords: operating systems, linux, virtual filesystem, inodes, hard links, permissions, setuid, umask, ext4, procfs

INTRODUCTION

Filesystems translate byte-oriented storage devices into hierarchical, human-navigable namespaces. Under the Unix philosophy, the abstraction extends beyond persistent media to expose device interfaces and kernel data structures through a common file interface.

Laboratory 5 explores this architecture across four core modules and four stretch problems:

a) **Filesystem Tree and Types**: Distinguishing regular files, directories, character devices, block devices, and symbolic links, alongside UsrMerge namespace organization.

b) **Inodes and Links**: Comparing directory entries with underlying inodes, hard links, dangling symbolic links, directory link count mechanics, and sparse file allocation.

c) **Permission Semantics**: Evaluating octal modes, the bitwise `umask` calculation, directory read versus execute semantics, setuid privilege elevation, and the sticky bit.

d) **Disks and Kernel Pseudo-Files**: Mapping block devices via `lsblk` and `df`, extracting hardware topology from `/proc/cpuinfo` and `/proc/meminfo`, and analyzing dynamic process resolution through `/proc/self`.

e) **Advanced System Inspections**: Header magic byte parsing, setuid auditing, unlinking open file descriptors, and measuring ext4 loopback inode table sizing.

1 THE TREE AND FILE TYPES

1.1 System Root and File Type Classification

In Unix-like systems, the first character of the `ls -l` output denotes the inode type. Inspecting `/bin`, `/dev/null`, `/etc`, `/etc/passwd`, and `/tmp` reveals five distinct filesystem abstractions:

```
$ ls -ld /etc /etc/passwd /bin /dev/null /tmp
lrwxrwxrwx   1 root root    7 Apr 22  2024 /bin -> usr/bin
crw-rw-rw-   1 root root 1, 3 Oct 12 15:33 /dev/null
drwxr-xr-x 107 root root 4096 Oct 12 15:38 /etc
-rw-r--r--   1 root root 2430 Oct 12 15:38 /etc/passwd
drwxrwxrwt  10 root root 4096 Oct 12 15:38 /tmp
```

Table 1.1 – Observed file types in system root

Path	Prefix	Filesystem Type	Semantic Role
`/bin`	`l`	Symbolic Link	Points to `usr/bin` (UsrMerge)
`/dev/null`	`c`	Character Device	Byte stream device (major 1, minor 3)
`/etc`	`d`	Directory	System-wide configuration table
`/etc/passwd`	`-`	Regular File	Plain text user database
`/tmp`	`d`	Directory (Sticky)	Shared workspace with mode 1777

1.2 The UsrMerge Architecture

On this system, `/bin` is a symbolic link pointing to `usr/bin`. Modern Debian and Ubuntu distributions implement UsrMerge to unify all system binaries under `/usr/bin`. This eliminates historical distinctions between essential single-user recovery binaries in `/bin` and multi-user packages in `/usr/bin`.

1.3 Device Files in /dev

Device files in `/dev` do not occupy disk data blocks; their inodes store major and minor device identifiers:

- **Character Device**: `/dev/null` (major 1, minor 3) streams bytes directly without intermediate caching.
- **Block Device**: `/dev/sda` (major 8, minor 0) and `/dev/sdd` (major 8, minor 48) support random block-level transfers backed by the page cache.

1.4 Filesystem Footprint: /var/log vs /usr

Running disk utilization measurements reveals:

```
$ du -sh /var/log 2>/dev/null ; du -sh /usr 2>/dev/null
422M    /var/log
7.4G    /usr
```

The `/usr` hierarchy occupies 7.4 GB, which is roughly 7.0 GB larger ($\approx 17.5\times$) than `/var/log` (422 MB). This disparity is expected because `/usr` houses shared libraries, system utilities, compilers, headers, and documentation, whereas `/var/log` only stores log text.

2 INODES, LINKS, AND SPACE ALLOCATION

2.1 Inode Metadata and Hard Links

A file's human-readable name is not stored within its inode. It exists solely as an entry in a parent directory mapping a string to an inode number.

```
$ echo hello > notes.txt
$ stat notes.txt
  File: notes.txt
  Size: 6           Blocks: 8          IO Block: 4096   regular file
Device: 8,48    Inode: 18829       Links: 1
```

Creating a hard link creates a second directory entry referencing the identical inode:

```
$ ln notes.txt hard.txt
$ ls -li notes.txt hard.txt
18829 -rw-r--r-- 2 ubuntu ubuntu 6 Oct 12 15:39 hard.txt
18829 -rw-r--r-- 2 ubuntu ubuntu 6 Oct 12 15:39 notes.txt
$ rm notes.txt
$ cat hard.txt
hello
```

The `rm` utility invokes `unlink()`, removing the name `notes.txt` from the directory and decrementing `i_nlink` from 2 to 1. Because `hard.txt` still points to inode 18829 and the link counter is positive, data blocks remain on disk.

2.2 Symbolic Link Mechanics

Unlike hard links, symbolic links are distinct inodes storing the path string of their target:

```
$ ln -s hard.txt soft.txt
$ ls -li hard.txt soft.txt
18829 -rw-r--r-- 1 ubuntu ubuntu 6 Oct 12 15:39 hard.txt
18830 lrwxrwxrwx 1 ubuntu ubuntu 8 Oct 12 15:39 soft.txt -> hard.txt
$ rm hard.txt
$ cat soft.txt
cat: soft.txt: No such file or directory
```

When `hard.txt` is removed, `soft.txt` becomes a dangling link. Opening it fails with `ENOENT`. The length of `soft.txt` is exactly 8 bytes because the path string `"hard.txt"` contains 8 characters.

2.3 Directory Link Count Arithmetic

Directories have fixed link counting rules:

```
$ mkdir d ; ls -ld d
drwxr-xr-x 2 ubuntu ubuntu 4096 Oct 12 15:39 d
$ mkdir d/x d/y ; ls -ld d
drwxr-xr-x 4 ubuntu ubuntu 4096 Oct 12 15:39 d
```

A new directory starts with 2 links: the parent entry `d` and the self-referencing entry `d/.`. When child directories `d/x` and `d/y` are created, each child contains a `..` entry pointing back to `d`, raising the link count to 4.

2.4 Logical Size versus Physical Allocation

File systems allocate storage in discrete blocks (4096 bytes in ext4):

```
$ echo x > one.txt ; ls -l one.txt ; du -h one.txt
-rw-r--r-- 1 ubuntu ubuntu 2 Oct 12 15:39 one.txt
4.0K    one.txt

$ truncate -s 1G big.img ; ls -lh big.img ; du -h big.img
-rw-r--r-- 1 ubuntu ubuntu 1.0G Oct 12 15:39 big.img
0       big.img
```

A 2-byte file requires one full 4 KiB filesystem block. Conversely, `truncate` creates a sparse file: it updates `i_size` in the inode metadata to 1 GiB without writing data blocks, resulting in 0 physical blocks allocated.

3 PERMISSIONS AND ACCESS CONTROL

3.1 Octal Modes and Evaluation Semantics

File permissions represent three 3-bit masks for User, Group, and Others:

- 600 (`rw-------`): Private SSH keys (`~/.ssh/id_rsa`).
- 644 (`rw-r--r--`): Web assets, documentation, `/etc/passwd`.
- 755 (`rwxr-xr-x`): Executables (`/bin/ls`) and public directories.
- 700 (`rwx------`): Private directories (`~/.ssh`).

When setting `chmod 000 s.txt`, reading fails even for the file's owner:

```
$ chmod 000 s.txt ; cat s.txt
cat: s.txt: Permission denied
```

The kernel checks whether `current_euid == inode->i_uid`. Since it matches, only owner bits are evaluated. Because owner bits are `0`, access is denied immediately. Ownership provides the authority to change mode bits with `chmod()`, not an automatic read bypass.

3.2 The umask Calculation

The observed `umask` is `0022`:

```
$ umask ; touch new.txt ; mkdir newdir ; ls -ld new.txt newdir
0022
-rw-r--r-- 1 ubuntu ubuntu    0 Oct 12 15:39 new.txt
drwxr-xr-x 2 ubuntu ubuntu 4096 Oct 12 15:39 newdir
```

Files start from base mode `0666`: $0666 \ \& \ \sim 0022 = 0644$ (`rw-r--r--`). Directories start from base mode `0777`: $0777 \ \& \ \sim 0022 = 0755$ (`rwxr-xr-x`).

3.3 Directory Read versus Execute Permissions

The `box` directory experiment demonstrates the separation between listing and traversing:

```
# Read-only directory
$ chmod u=r,go= box ; ls box ; cat box/f
f
cat: box/f: Permission denied


# Execute-only directory
```

```
$ chmod u=x,go= box ; ls box ; cat box/f
ls: cannot open directory 'box': Permission denied
hi
```

Read permission (`r`) allows reading the directory table, permitting `ls` to show the name `f`. However, traversing into the directory requires execute permission (`x`). Without `x`, path resolution fails. With execute-only permission (`x`), directory listing fails, but accessing known files directly succeeds.

3.4 Setuid and the Sticky Bit

```
$ ls -l /usr/bin/passwd ; ls -ld /tmp
-rwsr-xr-x 1 root root 64152 May 30  2024 /usr/bin/passwd
drwxrwxrwt 4 root root  4096 Oct 12 15:39 /tmp
```

The `s` in `/usr/bin/passwd` denotes setuid. When executed, the process assumes the effective UID of the file owner (`root`). This allows `passwd` to modify `/etc/shadow`, which normal users cannot write to. The `t` in `/tmp` represents the sticky bit: in a shared world-writable directory, users may only delete or rename files they own.

4 DISKS, TOPOLOGY, AND KERNEL PSEUDO-FILES

4.1 Disk Layout and Root Mount

```
$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sdc    8:32   0     5G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /

$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd       1007G  4.2G  952G   1% /

$ mount | grep " / "
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
```

The root filesystem resides on `/dev/sdd`, a 1 TB virtual SCSI disk formatted with ext4. It has 4.2 GB allocated out of 1007 GB (1% utilization). Swap is allocated on `/dev/sdc` (5 GB).

4.2 Hardware Introspection via /proc

```
$ grep -c processor /proc/cpuinfo
12
$ head -3 /proc/meminfo
MemTotal:       20323532 kB
MemFree:        17916984 kB
MemAvailable:   19695772 kB
```

The host exposes 12 logical CPU cores and 20,323,532 kB (≈ 19.38 GiB) of total RAM.

4.3 Self Process Resolution in /proc/self

```
$ head -5 /proc/self/status
Name:   head
Umask:  0022
State:  R (running)
```

The reported process name is `head`. In Linux, `/proc/self` is a dynamic symlink that resolves to the PID of the accessing process. Because `head` opens `/proc/self/status`, the kernel reports the status of `head`.

5 STRETCH EXPERIMENTS

5.1 (a) Magic Number Header Inspection

```
$ gzip -c /etc/passwd > notes.txt
$ xxd notes.txt | head -1
00000000: 1f8b 0808 d9d4 cc6a 0003 7061 7373 7764  .......j..passwd
```

The first two bytes, `0x1f 0x8b`, are the gzip magic number per RFC 1952. The `file` utility uses these bytes to identify the file type regardless of filename extension.

5.2 (b) Setuid Binary Audit

A system scan reveals essential setuid binaries:

```
/usr/bin/passwd, /usr/bin/su, /usr/bin/sudo, /usr/bin/mount, /usr/bin/umount,
/usr/lib/polkit-1/polkit-agent-helper-1, /usr/lib/openssh/ssh-keysign
```

5.3 (c) Unlinking an Open File Descriptor

```
$ exec 3< bigfile ; rm bigfile
$ ls -l /proc/$$/fd
lr-x------ 1 ubuntu ubuntu 64 Oct 12 15:40 3 -> /home/ubuntu/os-lab5/bigfile (deleted)
$ exec 3<&-
```

The link target indicates `(deleted)`. The directory entry is removed, but the inode and data blocks remain allocated until the last open descriptor is closed via `exec 3<&-`.

5.4 (d) Loopback ext4 Inode Allocation

```
$ truncate -s 64M disk.img ; mkfs.ext4 -F disk.img
Creating filesystem with 16384 4k blocks and 16384 inodes
$ sudo mount -o loop disk.img /mnt ; df -i /mnt
Filesystem     Inodes IUsed IFree IUse% Mounted on
/dev/loop0      16384    11 16373    1% /mnt
```

Formatting a 64 MiB disk image created exactly 16,384 inodes (one inode per 4 KiB block).

CONCLUSIONS

This laboratory verified core filesystem principles: directory names are decoupleable references to inodes; link counts govern inode lifecycle; file permissions separate directory enumeration from path traversal; and block allocation distinguishes logical length from physical storage.